

TIMERS Temporizadores y contadores

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- Introducción
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- Timer 1
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- Resumen



Introducción

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Los módulos Timer se utilizan principalmente para generar retardos, contar eventos. Estos módulos son un recurso que permiten liberar al procesador o CPU, lo que mejora la eficiencia del microcontrolador.

TIMER del PIC18F4550

- Determina intervalos de tiempo o cuenta los impulsos que provienen del exterior.
- Puede disponer de pre y/o pos escalador.
- **Existen dos modos de trabajo:**
 - como contador: la señal de reloj es externa o
 - como temporizador: la señal de reloj es interna
- **El microcontrolador PIC18F4550 contiene 4 modulo timer:**
 - Timer 0
 - Timer 1
 - Timer 2
 - Timer 3

11.0 TIMER0 MODULE

The Timer0 module incorporates the following features:

- Software selectable operation as a timer or counter in both 8-bit or 16-bit modes
- Readable and writable registers
- Dedicated 8-bit, software programmable prescaler
- Selectable clock source (internal or external)
- Edge select for external clock
- Interrupt on overflow

The T0CON register (Register 11-1) controls all aspects of the module's operation, including the prescale selection. It is both readable and writable.

A simplified block diagram of the Timer0 module in 8-bit mode is shown in Figure 11-1. Figure 11-2 shows a simplified block diagram of the Timer0 module in 16-bit mode.

- **Modo de trabajo:** Contador o Temporizador
- **Tamaño:** 8 o 16 bit
- **Registro de conteo:** TMR0L TMR0H (Leíble y escribible)
- **Conteo:** [0,255] ó [0,65535]
- **Pre-escalador:** 1,2,4,8,16,32,64,128,256.
- **Pos-escalador:** no tiene
- **Registro de configuración:** T0CON
- **Formula:** $\text{Tiempo} = (4/\text{Fosc}) * (N) * (\text{Pre-divisor})$
 - $N=256-\text{TMR0L}$
 - $N=65536-\text{TMR0L:TMR0H}$

REGISTER 11-1: T0CON: TIMER0 CONTROL REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
TMR0ON	T08BIT	T0CS	T0SE	PSA	T0PS2	T0PS1	T0PS0
bit 7							bit 0

- bit 7 **TMR0ON:** Timer0 On/Off Control bit
1 = Enables Timer0
0 = Stops Timer0
- bit 6 **T08BIT:** Timer0 8-Bit/16-Bit Control bit
1 = Timer0 is configured as an 8-bit timer/counter
0 = Timer0 is configured as a 16-bit timer/counter
- bit 5 **T0CS:** Timer0 Clock Source Select bit
1 = Transition on T0CKI pin
0 = Internal instruction cycle clock (CLKO)
- bit 4 **T0SE:** Timer0 Source Edge Select bit
1 = Increment on high-to-low transition on T0CKI pin
0 = Increment on low-to-high transition on T0CKI pin
- bit 3 **PSA:** Timer0 Prescaler Assignment bit
1 = Timer0 prescaler is NOT assigned. Timer0 clock input bypasses prescaler.
0 = Timer0 prescaler is assigned. Timer0 clock input comes from prescaler output.
- bit 2-0 **T0PS2:T0PS0:** Timer0 Prescaler Select bits
111 = 1:256 Prescale value
110 = 1:128 Prescale value
101 = 1:64 Prescale value
100 = 1:32 Prescale value
011 = 1:16 Prescale value
010 = 1:8 Prescale value
001 = 1:4 Prescale value
000 = 1:2 Prescale value

T0CON

TMR0ON	T08BIT	T0CS	T0SE	PSA	PS2	PS1	PS0
--------	--------	------	------	-----	-----	-----	-----

<7> TMR0ON: Encender/apagar Timer 0

0 = Detener

1 = Habilitar

<6> T08BIT: Tamaño

0 = 16 bit

1 = 8 bit

<5> T0CS: (Timer 0 Clock Source Select) Tipo de reloj para el Timer 0

0 = Los impulsos de reloj provienen del oscilador interno (Fosc/4) (Temporizador)

1 = Los impulsos de reloj provienen del exterior a través de la patita T0CKI(RA4) (Contador)

<4> T0SE: (Timer 0 Clock Edge Select) Elige el tipo de flanco activo de los impulsos externos en RA4/T0CKI

0 = Incremento del Timer0 cada flanco ascendente

1 = Incremento del Timer0 cada flanco descendente

<3> PSA: Asigna el divisor de frecuencias (pre escalador)

0 = Si

1 = No

<2-0> PS2-PS1-PS0: Seleccionan el rango del divisor de frecuencia

			Pre divisor TMR0	
PS2	PS1	PS0	PSA=0	PSA=1
0	0	0	1:2	1:1
0	0	1	1:4	1:1
0	1	0	1:8	1:1
0	1	1	1:16	1:1
1	0	0	1:32	1:1
1	0	1	1:64	1:1
1	1	0	1:128	1:1
1	1	1	1:256	1:1

TIMER 0

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*Registros asociados al TIMER 0***TABLE 11-1: REGISTERS ASSOCIATED WITH TIMER0**

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset Values on page
TMR0L	Timer0 Register Low Byte								54
TMR0H	Timer0 Register High Byte								54
INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF	53
INTCON2	$\overline{\text{RBPU}}$	INTEDG0	INTEDG1	INTEDG2	—	TMR0IP	—	RBIP	53
T0CON	TMR0ON	T08BIT	T0CS	T0SE	PSA	T0PS2	T0PS1	T0PS0	54
TRISA	—	TRISA6 ⁽¹⁾	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0	56

Diagrama del bloque del TIMER 0

FIGURE 11-1: TIMER0 BLOCK DIAGRAM (8-BIT MODE)

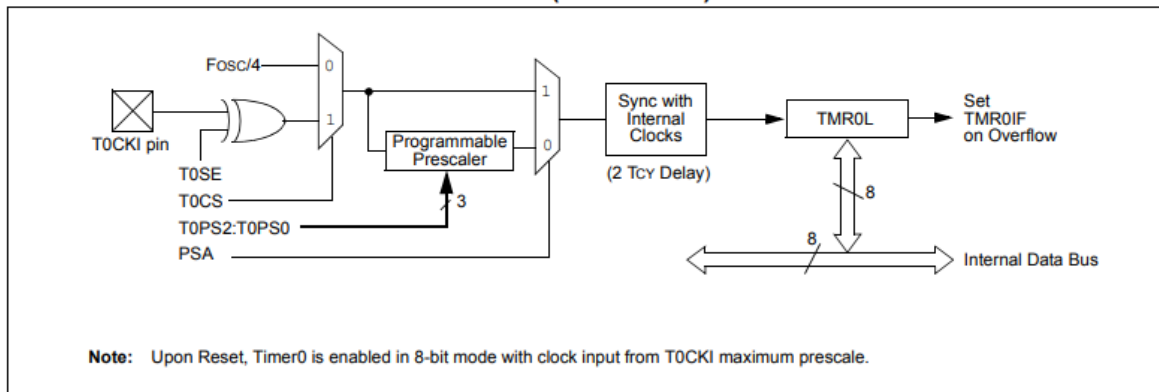
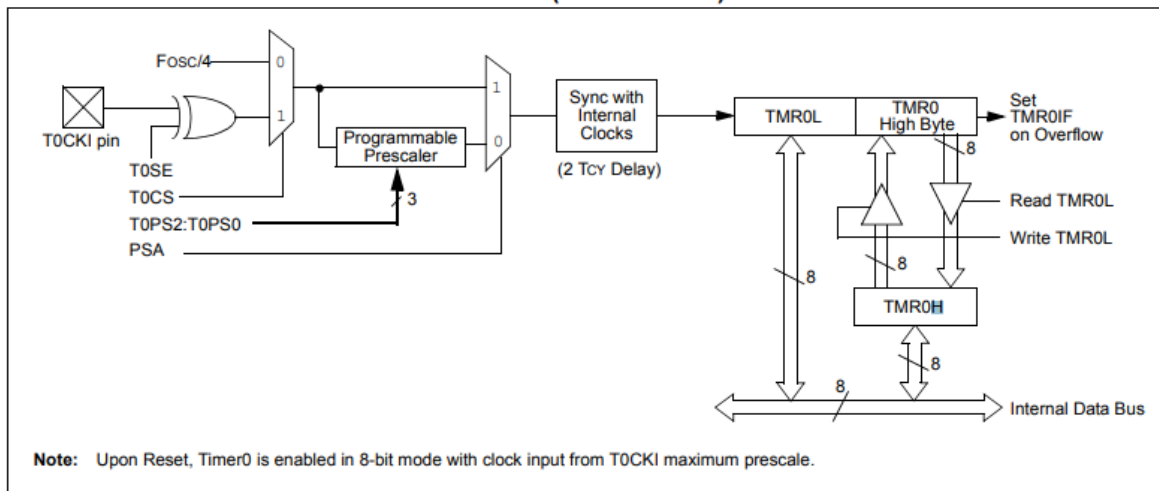


FIGURE 11-2: TIMER0 BLOCK DIAGRAM (16-BIT MODE)



TIMER 0

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Ejemplo 1: Calcular el tiempo máximo que se puede temporizar con el timer0 para 8 y 16 bit usando un cristal de 20Mhz.

Solución:

Para 8 bit

$$\begin{aligned}\text{Tiempo} &= (4/\text{Fosc}) * (256 - \text{TMR0L}) * (\text{Pre-divisor}) \\ &= (4/20\text{Mhz}) * (256 - 0) * (256) \\ &= \mathbf{13,1072\text{ms}}\end{aligned}$$

Para 16 bit

$$\begin{aligned}\text{Tiempo} &= (4/\text{Fosc}) * (65536 - \text{TMR0H} : \text{TMR0L}) * (\text{Pre-divisor}) \\ &= (4/20\text{Mhz}) * (65536 - 0) * (256) \\ &= \mathbf{3,35\text{ Seg}}\end{aligned}$$

TIMER 0

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Ejemplo 2:

- a) Calcule el valor de TMR0H:TMR0L para un tiempo de 2 seg usando un cristal de 20Mhz.
 b) escriba las subrutinas necesarias sin usar interrupciones.
 c) escriba las subrutinas necesarias usando interrupciones.

Solución, parte a:

Para 16 bit

$$\text{Tiempo} = (4/\text{Fosc}) * (65536 - \text{TMR0H:TMR0L}) * (\text{Pre-divisor})$$

$$2 \text{ seg} = (4/20\text{Mz}) * (65536 - \text{TMR0H:TMR0L}) * (256)$$

$$\text{TMR0H:TMR0L} = 65536 - \frac{2 \text{ Seg}}{(4/20\text{Mz}) * (256)}$$

$$\text{TMR0H:TMR0L} = 26473,5 \approx \mathbf{26474}$$

$$\mathbf{26474} = 0110011101101010$$

$$\text{TMR0H} = 01100111$$

$$\text{TMR0L} = 01101010$$

TIMER 0**Ejemplo 2:**

- a) Calcule el valor de `TMR0H:TMR0L` para un tiempo de 2 seg usando un cristal de 20Mhz.
 b) escriba las subrutinas necesarias sin usar interrupciones.
 c) escriba las subrutinas necesarias usando interrupciones.

Solución, parte b:

```

CONFIGURAR_TIMER_0
    MOVLW    B'00000111'
    MOVWF    T0CON
    RETURN

RETARDO_TMR0_2S
    MOVLW    B'01100111'
    MOVWF    TMR0H
    MOVLW    B'01101010'
    MOVWF    TMR0L

    BCF      INTCON, TMR0IF
    BSF      TMR0H, TMR0ON
ESPERA    BTFSS    INTCON, TMR0IF
          GOTO     ESPERA
          RETURN

PROGRAMA
    ;código
    CALL     RETARDO_TMR0_2S
    ;código
  
```

TABLE 11-1: REGISTERS ASSOCIATED WITH TIMER0

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset Values on page
TMR0L	Timer0 Register Low Byte								54
TMR0H	Timer0 Register High Byte								54
INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF	53
INTCON2	RBP $\overline{U}$	INTEDG0	INTEDG1	INTEDG2	—	TMR0IP	—	RBIP	53
T0CON	TMR0ON	T08BIT	T0CS	T0SE	PSA	T0PS2	T0PS1	T0PS0	54
TRISA	—	TRISA6 ⁽¹⁾	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0	56

TIMER 0**Ejemplo 2:**

- a) Calcule el valor de TMR0H:TMR0L para un tiempo de 2 seg usando un cristal de 20Mhz.
 b) escriba las subrutinas necesarias sin usar interrupciones.
 c) escriba las subrutinas necesarias usando interrupciones.

Solución, parte c:

```

ORG          00h
GOTO         INICIO
ORG          08h
GOTO         ATENDER_INTERRUPCIONES_H
ORG          18h
GOTO         ATENDER_INTERRUPCIONES_L

INICIO
    CALL     CONFIGURAR_PUERTOS
    CALL     CONFIGURAR_TIMER_0
    CALL     CONFIGURAR_INTERRUPCIONES
    ;otras configuraciones iniciales
    GOTO     PROGRAMA

CONFIGURAR_TIMER_0
    MOVLW    B'00000111'
    MOVWF    T0CON
    MOVLW    B'01100111'
    MOVWF    TMR0H
    MOVLW    B'01101010'
    MOVWF    TMR0L
    RETURN
  
```

```

CONFIGURAR_INTERRUPCIONES
    ; Habilitar la interrupción del timer 0
    RETURN

ATENDER_INTERRUPCIONES_H
    ;guardar contexto
    ;verificar la causa y atender interrupción del timer 0
    ;restaurar contexto
    RETFIE

ATENDER_INTERRUPCION_TIMER_0
    BCF      T0CON,TMR0ON
    BCF      INTCON,TMR0IF
    MOVLW    B'01100111'
    MOVWF    TMR0H
    MOVLW    B'01101010'
    MOVWF    TMR0L
    ;código en relación al timer
    GOTO     SALIR_INTERRUPCIONES

PROGRAMA
    ;código
    BSF      TMR0H,TMR0ON

    ;código
    END
  
```

12.0 TIMER1 MODULE

The Timer1 timer/counter module incorporates these features:

- Software selectable operation as a 16-bit timer or counter
- Readable and writable 8-bit registers (TMR1H and TMR1L)
- Selectable clock source (internal or external) with device clock or Timer1 oscillator internal options
- Interrupt on overflow
- Module Reset on CCP Special Event Trigger
- Device clock status flag (T1RUN)

A simplified block diagram of the Timer1 module is shown in Figure 12-1. A block diagram of the module's operation in Read/Write mode is shown in Figure 12-2.

The module incorporates its own low-power oscillator to provide an additional clocking option. The Timer1 oscillator can also be used as a low-power clock source for the microcontroller in power-managed operation.

Timer1 can also be used to provide Real-Time Clock (RTC) functionality to applications with only a minimal addition of external components and code overhead.

Timer1 is controlled through the T1CON Control register (Register 12-1). It also contains the Timer1 Oscillator Enable bit (T1OSCEN). Timer1 can be enabled or disabled by setting or clearing control bit, TMR1ON (T1CON<0>).

- **Modo de trabajo:** Como Contador o Temporizador.
- **Registro de conteo:** TMR1H y TMR1L (Leíble y escribible).
- **Tamaño:** 16 bit.
- **Conteo:** de 0 a 65535
- **Pre-escalador:** 1,2,4,8
- **Pos-escalador:** no tiene.
- **Registro de configuración:** T1CON
- **Formula:** $\text{Tiempo} = (4/\text{Fosc}) * (65536 - \text{TMR1H} : \text{TMR1L}) * (\text{Pre-divisor})$



TIMER 1

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REGISTER 12-1: T1CON: TIMER1 CONTROL REGISTER

R/W-0	R-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
RD16	T1RUN	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON
bit 7							bit 0

- bit 7 **RD16:** 16-Bit Read/Write Mode Enable bit
1 = Enables register read/write of Timer1 in one 16-bit operation
0 = Enables register read/write of Timer1 in two 8-bit operations
- bit 6 **T1RUN:** Timer1 System Clock Status bit
1 = Device clock is derived from Timer1 oscillator
0 = Device clock is derived from another source
- bit 5-4 **T1CKPS1:T1CKPS0:** Timer1 Input Clock Prescale Select bits
11 = 1:8 Prescale value
10 = 1:4 Prescale value
01 = 1:2 Prescale value
00 = 1:1 Prescale value
- bit 3 **T1OSCEN:** Timer1 Oscillator Enable bit
1 = Timer1 oscillator is enabled
0 = Timer1 oscillator is shut off
The oscillator inverter and feedback resistor are turned off to eliminate power drain.
- bit 2 **T1SYNC:** Timer1 External Clock Input Synchronization Select bit
When TMR1CS = 1:
1 = Do not synchronize external clock input
0 = Synchronize external clock input
When TMR1CS = 0:
This bit is ignored. Timer1 uses the internal clock when TMR1CS = 0.
- bit 1 **TMR1CS:** Timer1 Clock Source Select bit
1 = External clock from RC0/T1OSO/T13CKI pin (on the rising edge)
0 = Internal clock (Fosc/4)
- bit 0 **TMR1ON:** Timer1 On bit
1 = Enables Timer1
0 = Stops Timer1

Diagrama del bloque del TIMER 1

FIGURE 12-1: TIMER1 BLOCK DIAGRAM

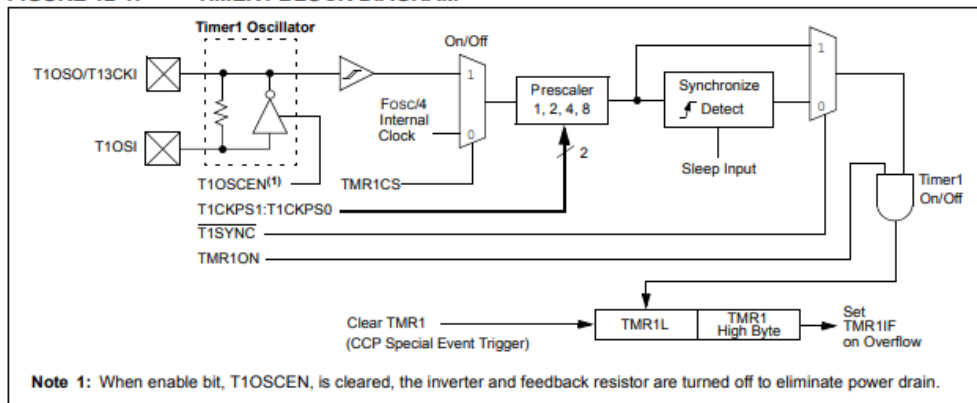
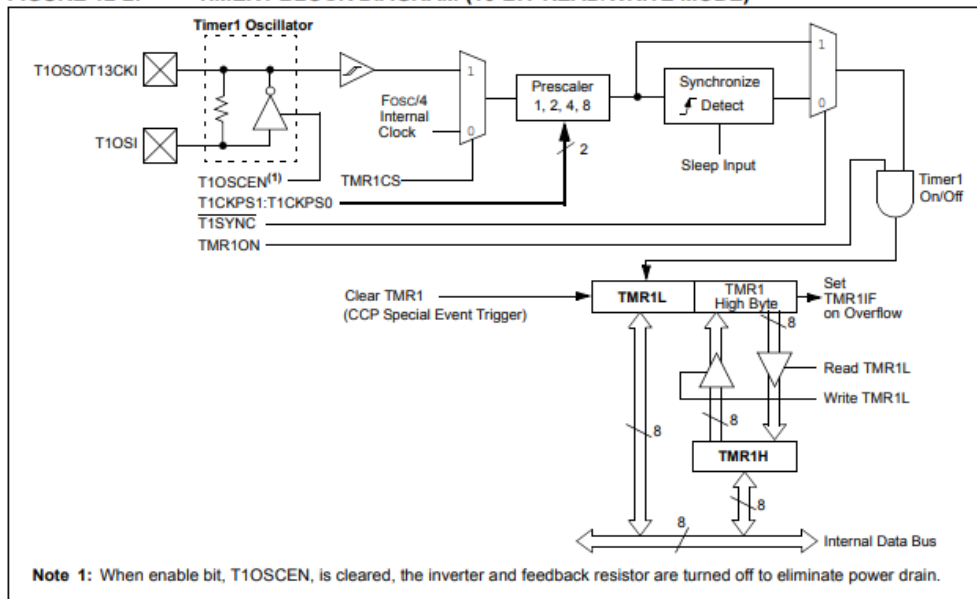


FIGURE 12-2: TIMER1 BLOCK DIAGRAM (16-BIT READ/WRITE MODE)



TIMER 1

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*Registros asociados al TIMER 1***TABLE 12-2: REGISTERS ASSOCIATED WITH TIMER1 AS A TIMER/COUNTER**

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset Values on page
INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF	53
PIR1	SPPIF ⁽¹⁾	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	56
PIE1	SPPIE ⁽¹⁾	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	56
IPR1	SPPIP ⁽¹⁾	ADIP	RCIP	TXIP	SSPIP	CCP1IP	TMR2IP	TMR1IP	56
TMR1L	Timer1 Register Low Byte								54
TMR1H	Timer1 Register High Byte								54
T1CON	RD16	T1RUN	T1CKPS1	T1CKPS0	T1OSCEN	T1SYN $\overline{C}$	TMR1CS	TMR1ON	54

Legend: — = unimplemented, read as '0'. Shaded cells are not used by the Timer1 module.

Note 1: These bits are unimplemented on 28-pin devices; always maintain these bits clear.

TIMER 1

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Ejemplo:

Calcular el tiempo máximo que se puede temporizar con el timer1 usando un cristal de 20Mhz

Solución:

$$\begin{aligned}\text{Tiempo} &= (4/\text{Fosc}) * (65536 - \text{TMR1H:TMR1L}) * (\text{Pre-divisor}) \\ &= (4/20\text{Mhz}) * (65536 - 0) * (8) \\ &= \mathbf{104,8576\text{ms}}\end{aligned}$$

TIMER 1

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Ejemplo: Calcular el valor de TMR1H:TMR1L para una duración de 1 segundo usando un oscilador externo de 32768hz con prescalador de 1

Solución:

$$\text{Tiempo} = \left(\frac{1}{F_{\text{osc}}} \right) * (65536 - \text{TMR1H:TMR1L}) * (\text{Pre-divisor})$$

$$\text{TMR1H:TMR1L} = 65536 - \frac{\text{Tiempo}}{(1/F_{\text{osc}}) * (\text{Pre-divisor})}$$

$$\text{TMR1H:TMR1L} = 65536 - \frac{1\text{seg}}{(1/32768\text{Hz}) * (1)}$$

$$\text{TMR1H:TMR1L} = 32768$$

$$\text{TMR1H:TMR1L} = 10000000:00000000$$

$$\text{TMR1H:TMR1L} = 80\text{H}:00\text{H}$$

TIMER 2

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13.0 TIMER2 MODULE

The Timer2 module timer incorporates the following features:

- 8-bit Timer and Period registers (TMR2 and PR2, respectively)
- Readable and writable (both registers)
- Software programmable prescaler (1:1, 1:4 and 1:16)
- Software programmable postscaler (1:1 through 1:16)
- Interrupt on TMR2 to PR2 match
- Optional use as the shift clock for the MSSP module

The module is controlled through the T2CON register (Register 13-1) which enables or disables the timer and configures the prescaler and postscaler. Timer2 can be shut off by clearing control bit, TMR2ON (T2CON<2>), to minimize power consumption.

A simplified block diagram of the module is shown in Figure 13-1.

13.1 Timer2 Operation

In normal operation, TMR2 is incremented from 00h on each clock ($F_{OSC}/4$). A 2-bit counter/prescaler on the clock input gives direct input, divide-by-4 and divide-by-16 prescale options. These are selected by the prescaler control bits, T2CKPS1:T2CKPS0 (T2CON<1:0>). The value of TMR2 is compared to that of the Period register, PR2, on each clock cycle. When the two values match, the comparator generates a match signal as the timer output. This signal also resets the value of TMR2 to 00h on the next cycle and drives the output counter/postscaler (see **Section 13.2 "Timer2 Interrupt"**).

The TMR2 and PR2 registers are both directly readable and writable. The TMR2 register is cleared on any device Reset, while the PR2 register initializes at FFh. Both the prescaler and postscaler counters are cleared on the following events:

- a write to the TMR2 register
- a write to the T2CON register
- any device Reset (Power-on Reset, $\overline{MCLR}$ Reset, Watchdog Timer Reset or Brown-out Reset)

TMR2 is not cleared when T2CON is written.

- **Modo de trabajo:** Como Temporizador.
- **Registro de conteo:** TMR2 (Leíble y escribible)
- **Tamaño:** 8 bit.
- **Conteo:** de 0 a 255
- **Pre-escalador:** 1,4,16
- **Pos-escalador:** 1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16
- **Registro de configuración:** T2CON
- **Formula:** $Tiempo = (4/F_{osc}) * (PR2+1) * (Pre-div) * (Pos-div)$



TIMER 2

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REGISTER 13-1: T2CON: TIMER2 CONTROL REGISTER

U-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0
—	T2OUTPS3	T2OUTPS2	T2OUTPS1	T2OUTPS0	TMR2ON	T2CKPS1	T2CKPS0
bit 7							bit 0

bit 7 **Unimplemented:** Read as '0'

bit 6-3 **T2OUTPS3:T2OUTPS0:** Timer2 Output Postscale Select bits

0000 = 1:1 Postscale

0001 = 1:2 Postscale

•

•

•

1111 = 1:16 Postscale

bit 2 **TMR2ON:** Timer2 On bit

1 = Timer2 is on

0 = Timer2 is off

bit 1-0 **T2CKPS1:T2CKPS0:** Timer2 Clock Prescale Select bits

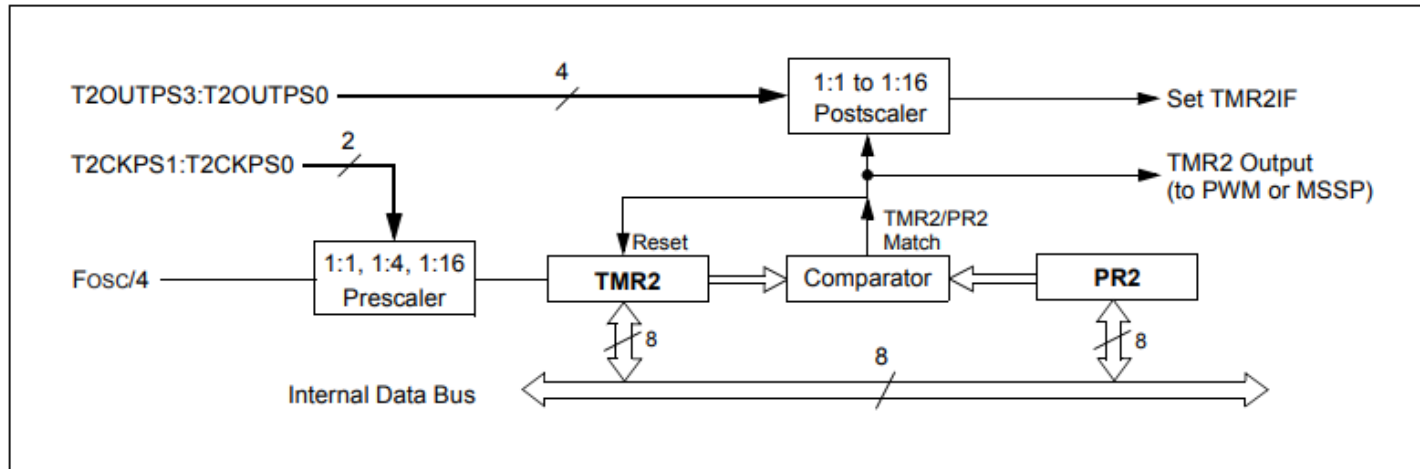
00 = Prescaler is 1

01 = Prescaler is 4

1x = Prescaler is 16

Diagrama del bloque del TIMER 2

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FIGURE 13-1: TIMER2 BLOCK DIAGRAM

TIMER 2

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*Registros asociados al TIMER 2***TABLE 13-1: REGISTERS ASSOCIATED WITH TIMER2 AS A TIMER/COUNTER**

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset Values on page
INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF	53
PIR1	SPPIF ⁽¹⁾	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	56
PIE1	SPPIE ⁽¹⁾	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	56
IPR1	SPPIP ⁽¹⁾	ADIP	RCIP	TXIP	SSPIP	CCP1IP	TMR2IP	TMR1IP	56
TMR2	Timer2 Register								54
T2CON	—	T2OUTPS3	T2OUTPS2	T2OUTPS1	T2OUTPS0	TMR2ON	T2CKPS1	T2CKPS0	54
PR2	Timer2 Period Register								54

TIMER 2

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Ejemplo: Calcular el tiempo máximo que se puede temporizar con el timer 2 usando un cristal de 20Mhz

Solución:

$$\begin{aligned}\text{Tiempo} &= (4/F_{\text{osc}}) * (PR2+1) * (\text{Pre-div}) * (\text{Pos-div}) \\ &= (4/20\text{Mhz}) * (255+1) * (16) * (16) \\ &= \mathbf{13,1072\text{ms}}\end{aligned}$$

TIMER 2

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Ejemplo: Utilizando timer 2 genere una señal de 50Hz con un cristal de 20Mhz

Solución: *Calculo de PR2*

$$T = \frac{1}{50Hz} \rightarrow T = 20mS$$

$$Tiempo = (4/Fosc) * (PR2 + 1) * (Pre - div) * (Pos - div)$$

$$10ms = (4/20Mhz) * (PR2 + 1) * (16) * (16)$$

$$PR2 = \frac{10mS}{(4/20Mhz) * (16) * (16)} - 1$$

$$PR2 = 194,3$$

$$PR2 = 194$$

TIMER 2

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Ejemplo: Utilizando timer 2 genere una señal de 50Hz con un cristal de 20Mhz

Solución: Código en lenguaje ensamblador

```
; PRE-D DE 16 Y UN POS-D DE 16 PARA 10MS PR2=194,3

CONFIGURAR_TIMER_2
    MOVLW    B'01111011'
    MOVWF    T2CON
    RETURN

TIMER2_10MS
    MOVLW    .194
    MOVWF    PR2

    BCF      PIR1,TMR2IF
    BSF      T2CON,2

ESPERA    BTFSS    PIR1,TMR2IF
           BRA      ESPERA
           BCF      T2CON,2
           RETURN
```

TABLE 13-1: REGISTERS ASSOCIATED WITH TIMER2 AS A TIMER/COUNTER

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset Values on page
INTCON	GIE/GIEH	PEIE/GIEL	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF	53
PIR1	SPPIF ⁽¹⁾	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	56
PIE1	SPPIE ⁽¹⁾	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	56
IPR1	SPPIp ⁽¹⁾	ADIP	RCIP	TXIP	SSPIP	CCP1IP	TMR2IP	TMR1IP	56
TMR2	Timer2 Register								54
T2CON	—	T2OUTPS3	T2OUTPS2	T2OUTPS1	T2OUTPS0	TMR2ON	T2CKPS1	T2CKPS0	54
PR2	Timer2 Period Register								54



BIBLIOGRAFÍA

- **Data Sheet:** PIC16F87X Microcontrollers.pdf
- **Libro:** 1ra parte Microcontroladores PIC - Diseño Práctico de Aplicaciones.pdf
- **Libro:** 2da Parte Microcontroladores PIC, Diseño Práctico de Aplicaciones 16F87x.pdf
- **Libro:** Fundamentos y Aplicaciones con PIC Valdez Pallas.pdf

Resumen

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- El microcontrolador PIC18F4550 dispone de 4 módulos timer.
- Cada timer tiene un registro de control, para configurarlo.
- El timer 0 se configura en T0CON, puede ser de 8 o 16 bit, ya sea como contador o temporizador, tiene pre escalador de 1,2,4...256.
- El timer 1 se configura en T1CON, es de 16 bit, ya sea como contador o temporizador, tiene pre escalador de 1,2,4,8.
- El timer 2 se configura en T2CON, es de 8 bit, solo temporizador, tiene pre escalador de 1,4,16 y pos escalador de 1,2,3..16.
- Cada timer se gestiona ya sea por interrupción o testeo de la bandera correspondiente.

Todo cambia cuando te das cuenta que ESTUDIAR es un acto de amor con tu "yo" del futuro porque es quien disfrutará de los frutos que hoy estas planeando.